

PROJECT REQUIREMENTS DOCUMENT

Software Requirements Specification

HRMS Platform

Human Resource Management System

Detailed feature specification for the development team

Build Stack: Node.js • React • MongoDB

Prepared for: Development Team

Version 1.0

1. Document Overview

Purpose. This document specifies the features and requirements for building the HRMS platform. It is written for the development team and describes what to build, how each feature should behave, and the standards the system must meet. It is a build specification, not a design exercise.

Product vision. A single, multi-tenant SaaS platform that manages the complete employee lifecycle — onboarding, employee records, attendance, leave, self-service, and (in later phases) payroll, performance, and analytics — comparable in scope to Keka, BambooHR, and Zoho People.

How to read this document. Section 6 details the Core modules to build first; each module lists its objective, features, user flow, roles, business rules, and acceptance criteria. Section 7 lists Premium (optional / paid-tier) modules for later phases. Sections 8-11 cover non-functional targets, data, integrations, and the delivery roadmap.

2. Product Goals & Success Criteria

Goals

- Provide a single source of truth for all employee data.
- Reduce HR workload through employee and manager self-service.
- Automate attendance and leave according to configurable policies.
- Scale to 50,000+ employees across multiple tenant organizations.
- Keep data secure, isolated per tenant, and auditable.

Success Criteria (what 'done' looks like for v1)

- HR can onboard an employee end-to-end and the record appears in the directory.
- Employees can punch in/out and apply for leave without contacting HR.
- Managers can approve requests inside the application.
- Reports return accurate headcount, attendance, and leave data.
- The system stays responsive during the morning attendance spike.

3. User Personas & Roles

The system supports four primary roles. Each role inherits the capabilities of the roles below it where it makes sense (e.g., a Manager is also an Employee).

- Employee — manages own profile, attendance, leave, payslips, and self-service requests.
- Reporting Manager — everything an Employee can do, plus team visibility and approvals.
- HR / Admin — full employee management, policy and role configuration, all approvals, and reports.
- Leadership / Management — organization-wide dashboards and analytics (mostly read-only).

Permission Overview

Capability	Employee	Manager	HR / Admin	Leadership
View & manage own data	✓	✓	✓	✓
Apply leave / mark attendance	✓	✓	✓	✓
Approve team requests	—	✓	✓	—
Manage employee records	—	—	✓	—
Configure policies & roles	—	—	✓	—
View org-wide analytics	—	Team only	✓	✓

4. Technology Stack

Build the platform on the following stack.

Layer	Technology & purpose
Frontend	React (single-page app) with role-based UI rendering.
Backend	Node.js (Express or NestJS) exposing REST APIs; JWT-based authentication.
Database	MongoDB — multi-tenant, indexed; every record carries a tenantId.
Cache	Redis — sessions, frequently-read data, and rate limiting. (Optional)
Async jobs	Queue (BullMQ / Kafka / SQS) — payroll runs, report exports, bulk notifications, device sync. (Optional)
File storage	S3-compatible object storage — documents, payslips, profile photos, exports.(Free Option You can Use)
Notifications	Email / SMS / push provider integration.

5. System Overview

- Multi-tenant SaaS — each client company is an isolated tenant running on shared infrastructure; all data is scoped by tenantId.
- Modular structure — each functional area (Auth, Employee, Attendance, Leave, etc.) is a self-contained module with its own data and APIs.
- Request flow — React client → API gateway / load balancer → backend services → MongoDB, with Redis for caching, a queue for heavy jobs, and object storage for files.
- Stateless backend — services hold no session state, allowing horizontal scaling behind a load balancer.

6. Core Modules (Required — Phase 1)

These modules form the minimum viable product and must be built first. Each is specified in detail below.

6.1 Authentication, RBAC & Multi-Tenancy

Secure access control with strict tenant isolation and role-based permissions.

Key Features

- Email/password login plus SSO (Google / Microsoft); optional SAML for enterprise tenants.

- JWT-based sessions with refresh tokens and configurable session timeout.
- Password policy (length, complexity, expiry), forgot/reset password, and account lockout after repeated failures.
- Optional multi-factor authentication (MFA) per tenant.
- Role-based access control with predefined roles (Employee, Manager, HR/Admin, Leadership) and custom roles with granular permissions.
- Tenant isolation — every request is scoped to a single tenant; cross-tenant access is impossible.
- Audit logging of logins and sensitive actions.

User Flow

1. User opens the app and is shown the login screen.
2. User submits credentials or signs in via SSO; the backend validates and issues a JWT scoped to the tenant and role.
3. The UI renders the menu and screens permitted for that role.
4. Every subsequent API call validates the token, tenant, and required permission.

Roles & Permissions

- HR/Admin: create and assign roles, manage users, configure auth policies.
- All users: authenticate and manage their own session.

Business Rules & Edge Cases

- All data access is scoped by tenantId — no exceptions.
- Accounts lock after 5 consecutive failed attempts for 15 minutes.
- Audit log entries are immutable.

Acceptance Criteria

- A user in Tenant A can never read or write Tenant B data.
- A non-admin cannot reach admin-only endpoints, even by direct URL/API call.
- Every login and sensitive action is recorded in the audit log.

6.2 Organization & Employee Management

A single source of truth for organization structure and complete employee records across the entire lifecycle.

Key Features

- Organization setup: company profile, business units, departments, teams, locations, designations, and grades.
- Employee details / master record:
 - Personal: name, date of birth, gender, photo, marital status, nationality.
 - Contact: personal & official email, phone, current & permanent address, emergency contact.
 - Employment: unique auto-generated employee ID, date of joining, employment type, department, designation, grade, location, reporting manager, assigned shift.
 - Bank & statutory: bank account & IFSC, PAN, Aadhaar/SSN, PF/ESI/UAN where applicable.
 - Professional: education, prior experience, skills, certifications.
 - Documents: ID proofs, offer letter, contracts, and certificates stored in object storage.

- Employee directory: searchable and filterable (by department, location, designation) with profile cards and an org-chart view.
- Reporting hierarchy: assign managers, visualize the org chart, and prevent circular reporting.
- Employee lifecycle: onboarding → confirmation (probation end) → transfer / promotion / role or department change → exit & offboarding (triggers final settlement).
- Bulk import and export of employees via CSV/Excel.

User Flow

1. HR creates an employee and fills in the master record.
2. HR assigns role, reporting manager, department, designation, and shift.
3. The employee receives an invite and completes self-service onboarding (remaining personal details and documents).
4. The employee becomes active and appears in the directory and org chart.

Roles & Permissions

- HR/Admin: create and edit all employee records and org structure.
- Manager: view team members and their basic details.
- Employee: view the directory and edit their own profile (sensitive fields require approval).

Business Rules & Edge Cases

- Employee ID is unique per tenant; duplicate records are not allowed.
- Reporting structures must prevent circular hierarchies.
- Exit records cannot be deleted — they are archived and retained for compliance.
- Edits to sensitive fields (bank, statutory) require approval and are audit-logged.

Acceptance Criteria

- HR can create a complete employee record and it immediately appears in the directory.
- The org chart accurately reflects the reporting structure.
- Attempting a circular manager assignment is blocked with a clear error.

6.3 Attendance Management

Capture and manage daily attendance according to configurable shift and policy rules. (A detailed Attendance PRD already exists — match that depth.)

Key Features

- Multi-mode capture: web punch, mobile punch with GPS geofencing, biometric device integration, and IP-restricted punch.
- Shifts & work patterns: fixed, flexible, and rotational shifts; weekly offs; per-employee or per-team assignment.
- Attendance rules engine: grace period, late-mark, half-day, short-leave, and overtime calculation.
- Regularization: employees request corrections for missed or incorrect punches, routed to the manager for approval.
- Holiday and weekly-off calendars per location.
- Real-time attendance dashboard and monthly muster/register.
- Overtime tracking and reporting.

User Flow

1. Employee punches in/out via an allowed mode.
2. The system validates location/IP and applies the assigned shift rules.
3. Attendance status is marked (present, late, half-day, absent, etc.).
4. Exceptions are raised for regularization and routed to the manager.
5. On approval, attendance is updated.

Roles & Permissions

- Employee: punch and raise regularization requests.
- Manager: approve regularizations and view team attendance.
- HR/Admin: configure shifts, rules, and calendars; view all attendance.

Business Rules & Edge Cases

- Duplicate punches from multiple devices are handled idempotently.
- GPS/IP validation is enforced per the tenant's policy.
- Regularization is allowed only within a configurable window (e.g., current and previous month).

Acceptance Criteria

- Punches produce the correct status per the assigned shift rules.
- A burst of simultaneous morning punches is processed without failures.
- An approved regularization correctly updates the attendance record.

6.4 Leave Management

Manage leave policies, balances, and request workflows, integrated with attendance and (later) payroll.

Key Features

- Leave types: casual, sick, earned/privileged, comp-off, maternity/paternity, loss-of-pay (LOP), and custom types.
- Leave policies: accrual rules and frequency, carry-forward limits, encashment, max consecutive days, notice period, and eligibility by grade/location.
- Real-time leave balance tracking per employee.
- Apply for leave (full-day, half-day, or hourly), view balance and history, and cancel/withdraw requests.
- Single or multi-level approval workflow with approver comments.
- Holiday calendar integration and weekend handling.
- Leave–attendance coupling: approved leave reflects in attendance; LOP days feed payroll.

User Flow

1. Employee applies for leave and the available balance is checked.
2. The request is routed to the configured approver(s).
3. Approver approves or rejects with a comment.
4. On approval, the balance and attendance are updated, and payroll is informed of any LOP.

Roles & Permissions

- Employee: apply for, view, and cancel leave.
- Manager: approve or reject team leave.
- HR/Admin: configure leave types and policies; override balances when needed.

Business Rules & Edge Cases

- Employees cannot apply beyond their available balance unless LOP is permitted.
- Overlapping leave requests are blocked.
- Carry-forward and encashment are applied per policy at year-end.

Acceptance Criteria

- Balance decreases on approval and is restored on cancellation.
- LOP days are passed to payroll.
- Overlapping requests are prevented with a clear message.

6.5 Self-Service (ESS & MSS)

Let employees and managers complete common HR tasks without HR intervention.

Key Features

- Employee Self-Service (ESS): view and update profile (with approval), apply for leave, view attendance, download payslips and documents, view holidays, and raise requests.
- Manager Self-Service (MSS): team dashboard, approve leave/attendance/regularization, view team attendance and leave, run team reports, and initiate transfers.
- Role-based dashboard with relevant widgets and quick actions.

Roles & Permissions

- Employee: ESS capabilities.
- Manager: MSS capabilities in addition to ESS.

Business Rules & Edge Cases

- Changes to sensitive profile fields require approval before taking effect.

Acceptance Criteria

- An employee can complete routine tasks (apply leave, update profile, download payslip) without contacting HR.
- A manager can act on all pending team requests from one place.

6.6 Workflow & Approvals

A configurable approval engine reused across all modules.

Key Features

- Configurable approval chains (single or multi-level) per request type.
- Conditional routing (e.g., leave longer than X days also needs HR approval).
- SLA-based auto-escalation and reminders.
- Delegation when an approver is unavailable.
- Full approval history and audit trail per request.

User Flow

1. A request is created in any module.
2. It is routed to approvers per the configured chain.
3. Approvers act; the status updates and notifications are sent.
4. If an approver misses the SLA, the request escalates automatically.

Business Rules & Edge Cases

- Every approval action is logged.
- Requests escalate after the configured SLA is breached.

Acceptance Criteria

- A request follows the configured chain exactly and escalates on SLA breach.

6.7 Notifications

Keep users informed across channels for relevant events.

Key Features

- Channels: email, in-app, and push.
- Event-driven: request submitted/approved/rejected, missed punch, pending-approval reminders.
- Templated messages and per-user notification preferences.
- Optional digest/summary notifications.

Business Rules & Edge Cases

- User notification preferences are respected.
- Critical/security notifications cannot be disabled.

Acceptance Criteria

- Relevant events generate timely notifications on the user's chosen channels.

6.8 Reporting & Dashboards

Role-based visibility into the workforce with exportable reports.

Key Features

- Role-based dashboards (employee, manager, HR, leadership).
- Standard reports: headcount, attendance summary, leave balance and usage, late/absent, overtime, and attrition.
- Filters (date range, department, location) and export to CSV/Excel/PDF.
- Scheduled report delivery by email.

User Flow

1. User selects a report and applies filters.
2. The report is generated (large exports run asynchronously).
3. The user views and/or exports the result.

Roles & Permissions

- Data is scoped by role: employees see their own, managers see their team, HR/Leadership see the organization.

Business Rules & Edge Cases

- All report data is scoped by role and tenant.

Acceptance Criteria

- Reports return accurate, role-scoped data and export correctly in each format.

7. Premium / Optional Modules (Later Phases)

These modules are typically offered as paid add-ons and are built after the core platform is stable.

7.1 Payroll & Compliance

Run payroll accurately and stay compliant with statutory requirements.

Key Features

- Salary structure setup (earnings, deductions, reimbursements).
- Pay runs with payslip generation and bank-file export.
- Statutory deductions (PF / ESI / TDS / PT) and reports.
- LOP integration from leave & attendance.

7.2 Performance Management

Drive and track employee performance.

Key Features

- Goals / OKRs / KRAs and review cycles.
- 360° feedback, ratings, and calibration.
- Performance history per employee.

7.3 Recruitment / ATS

Manage hiring from job posting to offer.

Key Features

- Job postings and candidate pipeline.
- Interview scheduling and feedback.
- Offer management and conversion to employee.

7.4 Onboarding & Document Management

Streamline joining formalities and document handling.

Key Features

- Onboarding checklists and task assignment.
- Document vault with e-sign and verification.
- Pre-boarding for new hires.

7.5 Expense & Reimbursement / Travel

Handle employee expenses and travel requests.

Key Features

- Expense claims with policy limits and approval.
- Travel requests and approvals.
- Integration with payroll/finance.

7.6 Learning & Development (LMS)

Deliver and track employee training.

Key Features

- Courses, assignments, and certifications.
- Training records and progress tracking.

7.7 Asset Management

Track company assets issued to employees.

Key Features

- Issue and return of laptops and devices.
- Asset lifecycle and assignment history.

7.8 AI & Analytics

Advisory intelligence layered on top of HR data. All AI outputs are advisory only, carry a confidence score, and require human approval for critical actions.

Key Features

- Smart employee search (natural-language directory queries).
- AI-generated employee summaries.
- Workforce planning recommendations and skill-gap analysis.
- Attrition prediction and burnout-risk flags.
- Attendance anomaly detection and smart regularization suggestions.
- Resume screening for recruitment.

7.9 Helpdesk / HR Tickets

Let employees raise and track HR queries.

Key Features

- Ticketing with SLA tracking.
- Knowledge base / FAQs.

7.10 Integrations

Connect the HRMS to external systems.

Key Features

- Biometric devices, SSO (SAML/OAuth), calendar.
- Accounting (Tally / QuickBooks) and chat (Slack / Teams).

7.11 Mobile App

Native mobile access to core features.

Key Features

- Attendance, leave, approvals, and notifications on mobile.

8. Non-Functional Requirements

These are build targets the system must meet, not optional extras.

Area	Requirement
Scalability	Support 50,000+ employees across multiple tenants; scale horizontally; absorb the morning attendance punch spike.
Performance	Typical API responses under ~300 ms at the 95th percentile; paginated lists; indexed queries.
Availability	Target high availability (e.g., 99.9%) via stateless services and database replica sets.
Security	Encryption in transit and at rest; server-side RBAC; OWASP protections; rate limiting; PII protection; audit trails.
Multi-tenancy	Strict data isolation by tenantId across the database, cache, and APIs.
Observability	Centralized logging, metrics, tracing, and alerting.
Backup & DR	Regular automated backups with defined recovery objectives (RPO/RTO).
Maintainability	Modular codebase and documented APIs.

9. Data Requirements

High-level data entities the system must store in MongoDB (the team designs the detailed schema):

- Core entities: tenants, users, roles, employees, departments, designations, locations, shifts.
- Attendance & leave: attendanceRecords, leaveTypes, leavePolicies, leaveRequests, leaveBalances, holidays.
- Cross-cutting: approvals, notifications, documents, auditLogs.
- Every record carries a tenantId; indexes are required on tenantId plus common query fields.
- Sensitive changes (employee, bank, role) must be captured in auditLogs.

10. Integration Requirements

- Biometric attendance devices (core attendance capture).
- Email / SMS provider for notifications.
- SSO providers (Google, Microsoft, SAML).
- Object storage for documents and files.
- Later phases: accounting (Tally/QuickBooks), calendar (Google/Outlook), and chat (Slack/Teams).

11. Delivery Roadmap

Phase	Scope
Phase 1 (MVP)	Auth/RBAC/Tenancy, Organization & Employee, Attendance, Leave, Self-Service, Workflow & Approvals, Notifications, basic Reporting.
Phase 2	Payroll & Compliance, Onboarding & Document Management, advanced Reporting, Mobile App.
Phase 3	Performance, Recruitment/ATS, AI & Analytics, LMS, Expense & Travel, Asset Management, Integrations, Helpdesk.

12. Assumptions & Dependencies

- HR/Admin users are available to configure organization structure and policies.
- Object storage and an email/SMS provider are provisioned.
- Biometric hardware is available where device-based attendance is used.
- Reliable internet connectivity for the cloud-hosted SaaS.

13. Open Questions

- Which countries' payroll and statutory compliance must v1 (or later) support?
- How deep should skill-inventory and workforce-planning data go?
- What are the photo/document storage and retention policies?
- Are there data-residency requirements per tenant or region?
- Which SSO providers are required for the first release?

— End of Requirements —